Making Comsic Pancakes!

Kshitish Kumar Ratha

And one of the most frustrating weeks of my 21 years existence.

Look out for these fun bits

Cosmic Webs

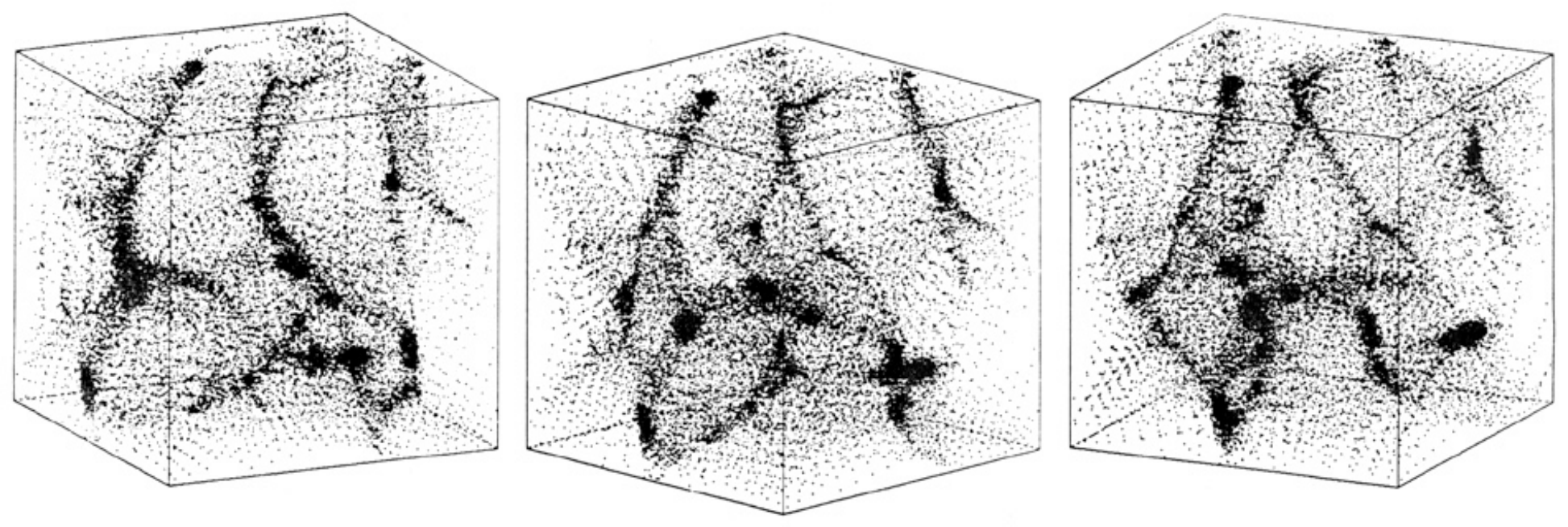


Figure 1: Cosmic Webs by Davis et al. [1]

The Paper That Started It All

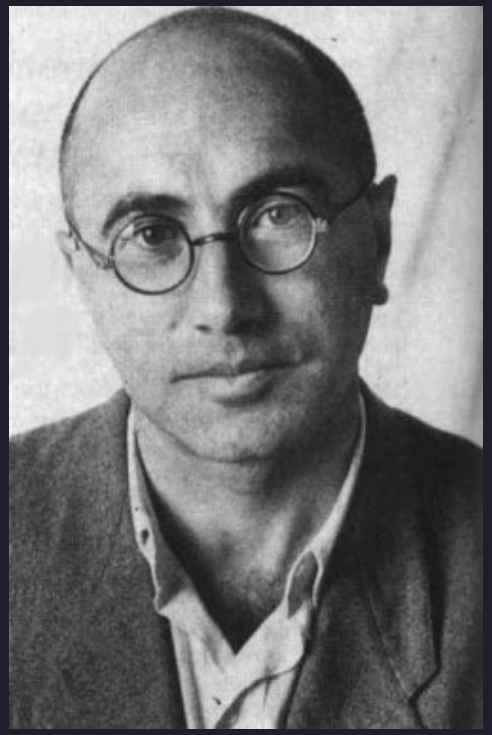
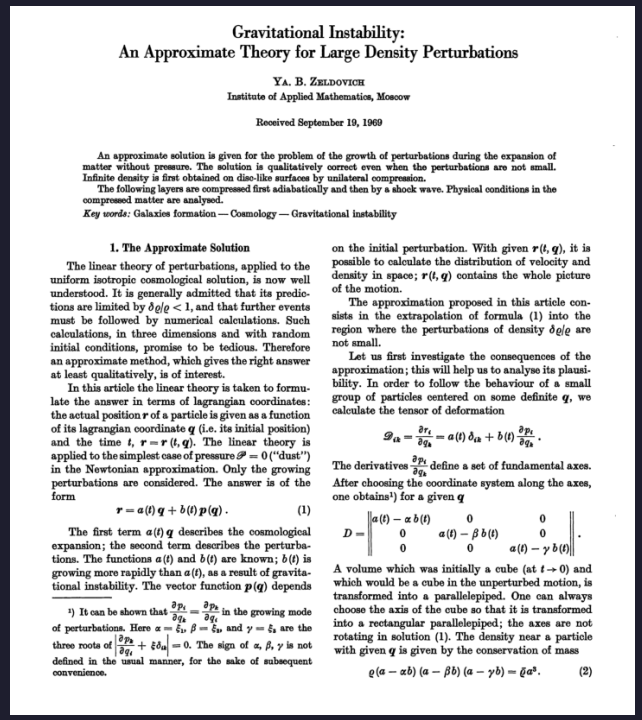


Figure 2: Yakov “Ya” Zel’dovich



Gravitational Instability:
An Approximate Theory for Large Density Perturbations

Ya. B. Zeldovich
Institute of Applied Mathematics, Moscow

Received September 19, 1969

An approximate solution is given for the problem of the growth of perturbations during the expansion of matter without pressure. The solution is qualitatively correct even when the perturbations are not small. Infinite density is first obtained on disc-like surfaces by unilateral compression.

The following layers are compressed first adiabatically and then by a shock wave. Physical conditions in the compressed matter are analysed.

Key words: Galaxies formation — Cosmology — Gravitational instability

1. The Approximate Solution

The linear theory of perturbations, applied to the uniform isotropic cosmological solution, is now well understood. It is generally admitted that its predictions are limited by $\delta\varrho/\varrho < 1$, and that further events must be followed by numerical calculations. Such calculations, in three dimensions and with random initial conditions, promise to be tedious. Therefore an approximate method, which gives the right answer at least qualitatively, is of interest.

In this article the linear theory is taken to formulate the answer in terms of lagrangian coordinates: the actual position $\boldsymbol{r}$ of a particle is given as a function of its lagrangian coordinate $\boldsymbol{q}$ (i.e. its initial position) and the time t, $\boldsymbol{r} = \boldsymbol{r}(t, \boldsymbol{q})$. The linear theory is applied to the simplest case of pressure $\mathscr{P} = 0$ (“dust”) in the Newtonian approximation. Only the growing perturbations are considered. The answer is of the form

$$\boldsymbol{r} = a(t)\,\boldsymbol{q} + b(t)\,\boldsymbol{p}(\boldsymbol{q})\,. \qquad (1)$$

The first term $a(t)\,\boldsymbol{q}$ describes the cosmological expansion; the second term describes the perturbations. The functions $a(t)$ and $b(t)$ are known; $b(t)$ is growing more rapidly than $a(t)$, as a result of gravitational instability. The vector function $\boldsymbol{p}(\boldsymbol{q})$ depends on the initial perturbation. With given $\boldsymbol{r}(t, \boldsymbol{q})$, it is possible to calculate the distribution of velocity and density in space; $\boldsymbol{r}(t, \boldsymbol{q})$ contains the whole picture of the motion.

The approximation proposed in this article consists in the extrapolation of formula (1) into the region where the perturbations of density $\delta\varrho/\varrho$ are not small.

Let us first investigate the consequences of the approximation; this will help us to analyse its plausibility. In order to follow the behaviour of a small group of particles centered on some definite $\boldsymbol{q}$, we calculate the tensor of deformation

$$\mathscr{D}_{ik} = \frac{\partial r_i}{\partial q_k} = a(t)\,\delta_{ik} + b(t)\,\frac{\partial p_i}{\partial q_k}\,.$$

The derivatives $\frac{\partial p_i}{\partial q_k}$ define a set of fundamental axes. After choosing the coordinate system along the axes, one obtains[1]) for a given $\boldsymbol{q}$

$$D = \begin{Vmatrix} a(t) - \alpha\, b(t) & 0 & 0 \\ 0 & a(t) - \beta\, b(t) & 0 \\ 0 & 0 & a(t) - \gamma\, b(t) \end{Vmatrix}.$$

A volume which was initially a cube (at $t \to 0$) and which would be a cube in the unperturbed motion, is transformed into a parallelepiped. One can always choose the axis of the cube so that it is transformed into a rectangular parallelepiped; the axes are not rotating in solution (1). The density near a particle with given $\boldsymbol{q}$ is given by the conservation of mass

$$\varrho(a - \alpha b)\,(a - \beta b)\,(a - \gamma b) = \bar{\varrho} a^3. \qquad (2)$$

[1]) It can be shown that $\frac{\partial p_i}{\partial q_k} = \frac{\partial p_k}{\partial q_i}$ in the growing mode of perturbations. Here $\alpha = \xi_1$, $\beta = \xi_2$, and $\gamma = \xi_3$ are the three roots of $\left|\frac{\partial p_k}{\partial q_i} + \xi\delta_{ik}\right| = 0$. The sign of α, β, γ is not defined in the usual manner, for the sake of subsequent convenience.

Figure 3: Paper from 1969

Original Form of Zel'dovich Approximation [2]

$$\boldsymbol{r} = a(t)\boldsymbol{q} + b(t)\boldsymbol{p}(\boldsymbol{q})$$

- $a(t)\boldsymbol{q}$: Describes Cosmological Expansion
- $b(t)$ grows more rapidly than $a(t)$ which causes gravitational instability.
- Extrapolate the above formula in regions where perturbations $\frac{\delta\rho}{\rho}$ are not small.
- $$a(Z) = \frac{1}{1+Z} \quad b(Z) = \frac{2}{5}\frac{1}{(1+Z)^2}$$

Mentions the formula in appendix and refuses to elaborate about b(t)

Connection with Caustics [3]

$$\boldsymbol{x} = \boldsymbol{q} + \tau \boldsymbol{v}(\boldsymbol{q})$$

- Spectrum of density perturbations $\longrightarrow$ affects the formation of structure $\longrightarrow$ top-down or bottom-up scenario.
- Collapse of matter into caustics which take the form of sheets, filaments and nodes.
- The displacement field $\boldsymbol{v}$ can be complex.
- Rotations difficult to see because of expansion.

$$\boldsymbol{v} = \nabla \Phi$$

Cosmic Pancakes ?!

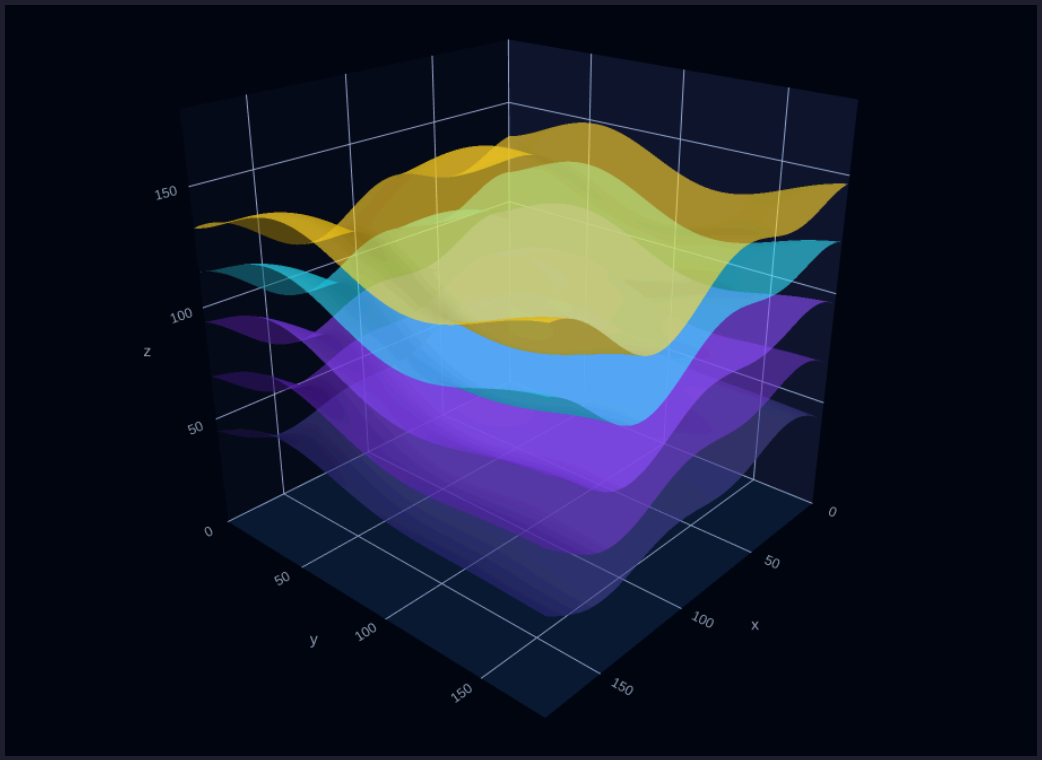


Figure 4: Pancakes in my head…

Seeing Zel'dovich Approximation in Action

Compared to the general simulations and computations that physicists do, calculation for Zel'dovich approximation is trivial.

Steps

1. Generate a gaussian random field for density δ.
2. Calculate the change in position $\boldsymbol{v}$ using the formula $\boldsymbol{v} = \nabla\Phi$ where Φ is the potential related to δ.
3. Assume all particles started uniformly spaced at $t = 0$.
4. Calculate the new position of particles using $\boldsymbol{x} = \boldsymbol{q} + \tau\boldsymbol{v}(\boldsymbol{q})$ where τ is some constant.

stonks?!!

Not so fast

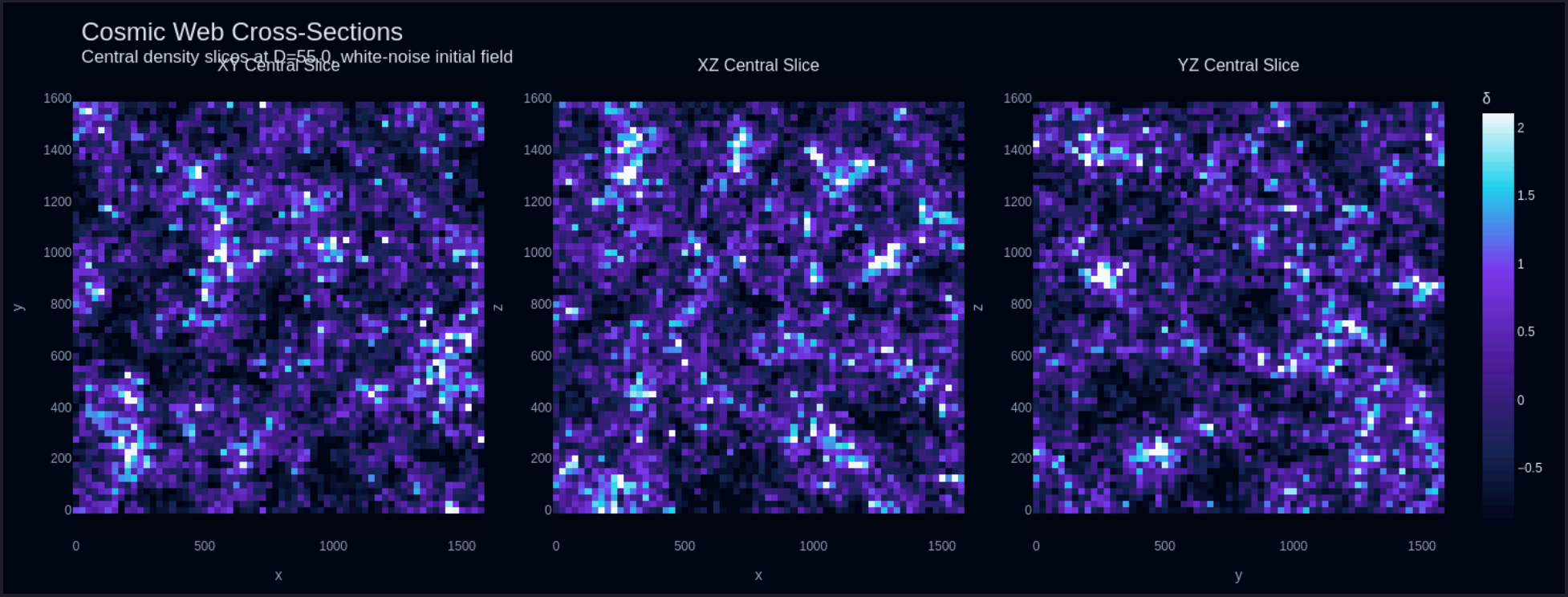


Figure 5: Gaussian Random Delta *(Click Me)*

Uhhh.. Did it work? [4]

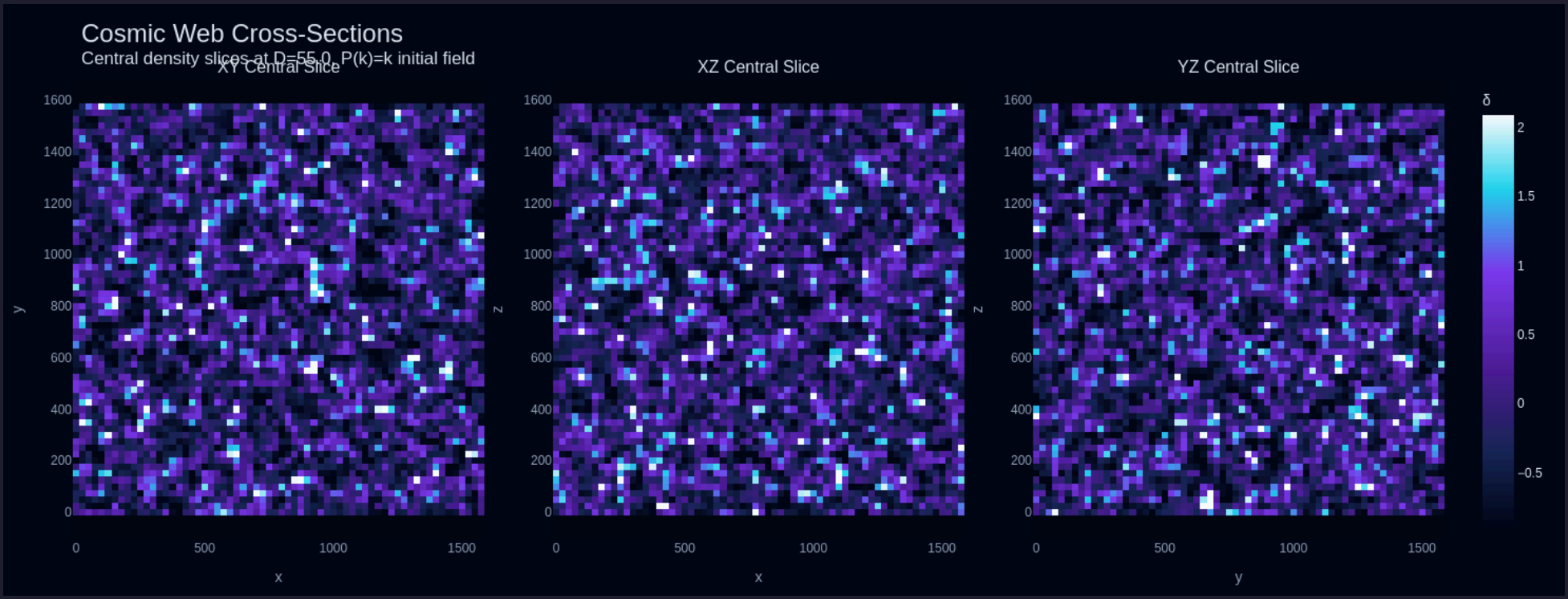


Figure 6: Esienstein Form *(Click Me)*

Magic Spectrum

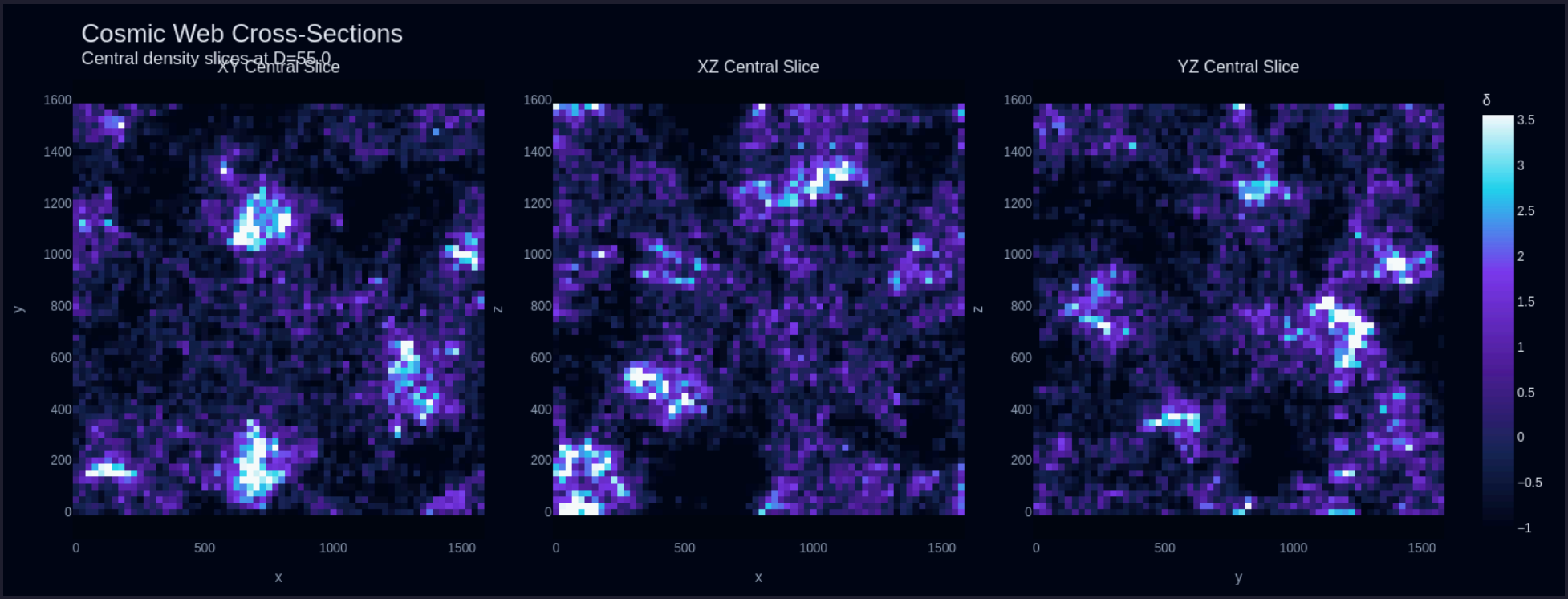


Figure 7: Some visible structure? *(Click Me)*

Where is my pancake!!!!

- Is the power spectrum still not working?
- Do I really need to go through old CAMB code?
- Should I ask Prof. Bagla?
- Do I bite the bullet and read the Esienstein and Hu paper?
- Should I keep falling prey to Sunk Cost Fallacy?

Are pancakes even real?

The Search For Pancakes [5]

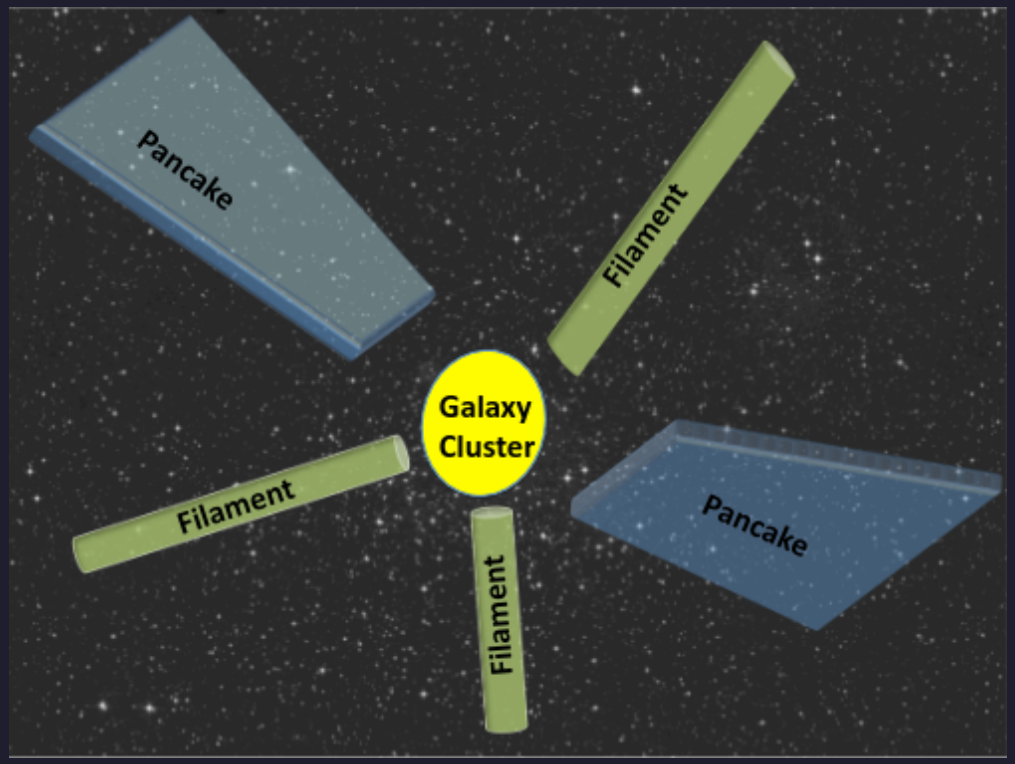


Figure 8: This is a pancake?

Maybe Pancakes Are The Blobs We Made Along The Way

- Void fraction: 0.2078399658203125
- Pancake fraction: 0.5223960876464844
- Filament fraction:0.2500762939453125
- Knot fraction: 0.019687652587890625

Does my code compile?	Yes
Does my code work?	Probably
Is the underlying physics correct?	Who Knows?
Are these Zel’dovich’s pancakes?	I don’t think so

References

[1] “Zel’dovich Pancake Day! | In the Dark.” Accessed: Apr. 24, 2026. [Online]. Available: *https://telescoper.blog/2021/02/16/zeldovich-pancake-day/*

[2] Y. B. Zel'dovich, “Gravitational instability: An approximate theory for large density perturbations.,” *Astronomy and Astrophysics*, vol. 5, pp. 84–89, Mar. 1970, Accessed: Apr. 25, 2026. [Online]. Available: *https://ui.adsabs.harvard.edu/abs/1970A&A.....5...84Z/abstract*

[3] S. F. Shandarin and Y. B. Zeldovich, “The large-scale structure of the universe: Turbulence, intermittency,

structures in a self-gravitating medium," *Reviews of Modern Physics*, 1989, doi: *10.1103/revmodphys.61.185*.

[4] D. J. Eisenstein and W. Hu, "Power Spectra for Cold Dark Matter and its Variants," *The Astrophysical Journal*, vol. 511, no. 1, pp. 5–15, Jan. 1999, doi: *10.1086/306640*.

[5] M. O. Lindholmer and S. H. Hansen, "A Catalogue of Zeldovich Pancakes."